

8	(a) (i) energy = $5.75 \times 1.6 \times 10^{-13}$ = $9.2 \times 10^{-13} \text{ J}$	A1	[1]
	(ii) number = $1900 / (9.2 \times 10^{-13} \times 0.24)$ = $8.6 \times 10^{15} \text{ s}^{-1}$	C1	
		A1	[2]
	(b) (i) decay constant = $0.693 / (2.8 \times 365 \times 24 \times 3600)$ = $7.85 \times 10^{-9} \text{ s}^{-1}$ (allow 7.8 or 7.9 to 2 s.f.)	C1	
		A1	[2]
	(ii) $A = \lambda N$ $8.6 \times 10^{15} = 7.85 \times 10^{-9} \times N$ $N = 1.096 \times 10^{24}$	C1	
		C1	
	mass = $(1.096 \times 10^{24} \times 236) / (6.02 \times 10^{23})$ = 430 g	M1	
		A1	[4]
	(c) $0.84 = 1.9 \exp(-7.85 \times 10^{-9} t)$ $t = 1.04 \times 10^8 \text{ s}$ = 3.3 years	C1	
		A1	[2]

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Section B